

SQL Final Project Technical Report

Team D9

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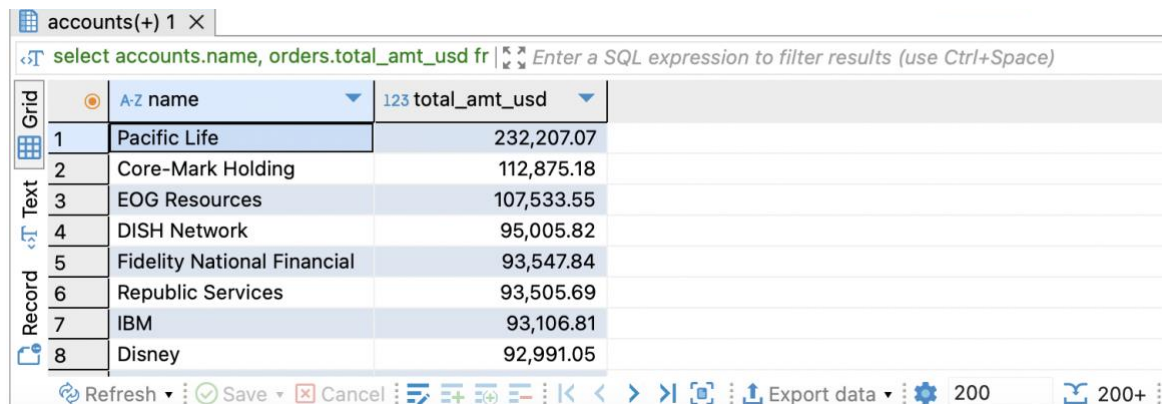
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--Q1

What is the current state of the company?

----Customer Information--

```
select accounts.name, orders.total_amt_usd
from orders inner join accounts
on accounts.id = orders.account_id
order by total_amt_usd
desc;
```



	A-Z name	123 total_amt_usd
1	Pacific Life	232,207.07
2	Core-Mark Holding	112,875.18
3	EOG Resources	107,533.55
4	DISH Network	95,005.82
5	Fidelity National Financial	93,547.84
6	Republic Services	93,505.69
7	IBM	93,106.81
8	Disney	92,991.05

----Well Known Customers--

select name from accounts;

	A-Z name	123 total_sales_revenue
1	Walmart	
2	Exxon Mobil	
3	Apple	
4	Berkshire Hathaway	
5	Mattel	
6	McKesson	
7	UnitedHealth Group	
8	CVS Health	

----Total Number of Customers--

select count(distinct(id)) from accounts;

select distinct(accounts.name),

sum(orders.standard_amt_usd) as *Total_Sales_Revenue*

from accounts inner join orders on accounts.id = orders.account_id

group by accounts.name

order by *Total_Sales_Revenue* desc ;

	A-Z name	123 total_sales_revenue
1	Core-Mark Holding	207,668.83
2	Thermo Fisher Scientific	116,371.79
3	BlackRock	114,450.64
4	Walgreens Boots Alliance	110,274.01
5	Citigroup	109,495.57
6	American Family Insurance Group	107,025.52
7	Cigna	106,865.84
8	ADP	106,177.22

select * from orders;

select * from region r ;

select * from accounts a ;

select * from sales_reps sr ;

select distinct(region_id) from sales_reps;

select accounts.name, orders.standard_amt_usd from accounts inner join orders

on accounts.id = orders.account_id order by orders.standard_amt_usd desc ;

accounts(+) 1 X

select accounts.name, orders.standard_amt_usd

	A-Z name	123 standard_amt_usd
1	Core-Mark Holding	112,729.09
2	State Farm Insurance Cos.	78,088.51
3	Kohl's	36,751.35
4	Travelers Cos.	35,344.17
5	Cigna	30,154.57
6	Kohl's	22,809.29
7	Thermo Fisher Scientific	22,085.74
8	BB&T Corp.	21,766.38

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----Sales Info---

----Total Sales Revenue in USD + Total Quantities Sold of Standard, Gloss, and Poster Paper--

```
select sum(orders.total_amt_usd) as Total_Revenue,
sum(orders.standard_qty) as Total_Standard_Paper,
sum(orders.gloss_qty) as Total_Gloss_Paper,
sum(orders.poster_qty) as Total_Poster_Paper
from orders;
```

Results 1 X

select sum(orders.total_amt_usd) as Total_Rev

	123 total_revenue	123 total_standard_paper	123 total_gloss_paper	123 total_poster_paper
1	23,141,511.83	1,938,346	1,013,773	723,646

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----First Sales Date--

```
select min(orders.occurred_at) as First_Sale_Date,
count(distinct (accounts.id)) as Total_Customers
from orders inner join accounts on orders.account_id = accounts.id;
```

Results 1 X

select min(orders.occurred_at) as First_Sale_C [Enter a SQL expression to filter results (use Ctrl+Space)]

	first_sale_date	123 total_customers
1	2013-12-04 04:22:44.000	350

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----Employee Info--

--Where Employees Operate--

select region.name as *Region* , count(sales_reps.region_id) as *Num_Employees*

from region

inner join sales_reps on region.id = sales_reps.region_id

group by region.name, sales_reps.region_id;

region 1 X

select region.name as Region , count(sales_reps.region_id) as Num_Employees [Enter a SQL expression to filter results (use Ctrl+Space)]

	A-Z region	123 num_employees
1	West	10
2	International	1
3	Midwest	9
4	Northeast	21
5	Southeast	10

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----Total Number of Employees--

select sum(standard_amt_usd) from orders o ;

select * from orders;

Results 1 X

select sum(standard_amt_usd) from orders o [Enter a SQL expression to filter results (use Ctrl+Space)]

	123 sum
1	9,672,346.54

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--Q2

Look into revenue streams

---- Total Revenue by Product Type

```
select sum(standard_amt_usd) as standard_revenue, sum(gloss_amt_usd) as gloss_revenue, sum(posters_amt_usd) as poster_revenue
from orders;
```

	123 standard_revenue	123 gloss_revenue	123 poster_revenue
1	9,672,346.54	7,593,159.77	5,876,005.52

---- Product Popularity by Order Count

```
select sum(standard_qty) as standard_orders, sum(gloss_qty) as gloss_orders, sum(posters_qty) as poster_orders
from orders;
```

	123 standard_orders	123 gloss_orders	123 poster_orders
1	1,938,346	1,013,773	723,646

----Monthly Revenue Trend

```
select to_char(occurred_at, 'YYYY-MM') as month, sum(total_amt_usd) as monthly_revenue
from orders
group by month
order by month;
```

Results 1 X

select to_char(occurred_at, 'YYYY-MM') as mo | Enter a SQL expression to filter results (use Ctrl+Space)

Grid	A-Z month	123 monthly_revenue
1	2013-12	377,331
2	2014-01	286,140.27
3	2014-02	349,721.34
4	2014-03	341,512.32
5	2014-04	344,893.99
6	2014-05	319,210.4
7	2014-06	297,655.65
8	2014-07	289,128.19

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----Monthly Revenue Trend by Product Type

```
select to_char(occurred_at, 'YYYY-MM') as month, sum(total_amt_usd) as
monthly_revenue,
sum(standard_amt_usd) as standard_revenue,
sum(gloss_amt_usd) as gloss_revenue,
sum.poster_amt_usd) as poster_revenue
from orders
group by month
order by month;
```

Results 1 X

select to_char(occurred_at, 'YYYY-MM') as mo | Enter a SQL expression to filter results (use Ctrl+Space)

Grid	A-Z month	123 monthly_	123 standard_rev	123 gloss_rever	123 poster_re
1	2013-12	377,331	132,504.46	154,938.14	89,888.4
2	2014-01	286,140.27	133,452.56	97,512.31	55,175.4
3	2014-02	349,721.34	142,010.41	118,918.73	88,792.2
4	2014-03	341,512.32	135,553.35	115,323.53	90,635.44
5	2014-04	344,893.99	131,686.1	121,914.73	91,293.16
6	2014-05	319,210.4	143,572.28	96,411.28	79,226.84
7	2014-06	297,655.65	140,917.6	94,441.41	62,296.64
8	2014-07	289,128.19	149,076.25	87,677.94	52,374

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----Revenue Contribution by Product Type for Sample Orders

```
select id as order_id,
(standard_amt_usd / total_amt_usd) * 100 as standard_pct,
(gloss_amt_usd / total_amt_usd) * 100 as gloss_pct,
.poster_amt_usd / total_amt_usd) * 100 as poster_pct
from orders
limit 30;
```

orders 1 X

select id as order_id, (standard_amt_usd / total) Enter a SQL expression to filter results (use Ctrl+Space)

	123 order_id	123 standard_pct	123 gloss_pct	123 poster_pct
1	1	63.0522996004	16.9277708721	20.0199295275
2	2	55.1852994418	17.8745423537	26.9401582045
3	3	54.6458295756	45.3541704244	0
4	4	74.9874770412	25.0125229588	0
5	5	54.7966934082	22.0856338143	23.1176727776
6	6	48.1583509019	16.8432888264	34.9983602717
7	7	33.6397009745	16.4977973568	49.8625016687
8	8	23.0996004288	17.1537861807	59.7466133905

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--Q3

Is this business growing?

select

```

date_trunc('month', occurred_at) as month,
sum(standard_qty) as standard_total_qty,
sum(gloss_qty) as gloss_total_qty,
sum(poster_qty) as paper_total_qty,
sum(standard_qty + gloss_qty + poster_qty) as total_qty,
sum(standard_amt_usd) as standard_total_amt_usd,
sum(gloss_amt_usd) as gloss_total_amt_usd,
sum(poster_amt_usd) as paper_total_amt_usd,
sum(total_amt_usd) as total_amt_usd

```

from orders

group by month

order by month;

Results 1 X

select date_trunc('month', occurred_at) as mo Enter a SQL expression to filter results (use Ctrl+Space)

	month	123 standarc	123 gloss_t	123 paper	123 total_q	123 standarc	123 gloss_	123 paper	123 total_an
1	2013-12-01 00:00:00.000	26,554	20,686	11,070	58,310	132,504.46	154,938.14	89,888.4	377,331
2	2014-01-01 00:00:00.000	26,744	13,019	6,795	46,558	133,452.56	97,512.31	55,175.4	286,140.27
3	2014-02-01 00:00:00.000	28,459	15,877	10,935	55,271	142,010.41	118,918.73	88,792.2	349,721.34
4	2014-03-01 00:00:00.000	27,165	15,397	11,162	53,724	135,553.35	115,323.53	90,635.44	341,512.32
5	2014-04-01 00:00:00.000	26,390	16,277	11,243	53,910	131,686.1	121,914.73	91,293.16	344,893.99
6	2014-05-01 00:00:00.000	28,772	12,872	9,757	51,401	143,572.28	96,411.28	79,226.84	319,210.4
7	2014-06-01 00:00:00.000	28,240	12,609	7,672	48,521	140,917.6	94,441.41	62,296.64	297,655.65
8	2014-07-01 00:00:00.000	29,875	11,706	6,450	48,031	149,076.25	87,677.94	52,374	289,128.19

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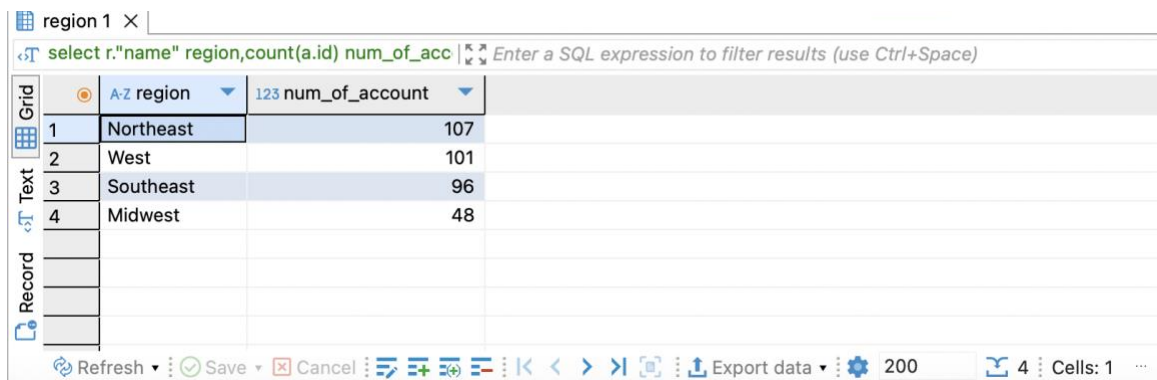
--Q4

Could we increase efficiency?

--4a

----Identify how many accounts are allocated for each region

```
select r."name" region, count(a.id) num_of_account from accounts a
join sales_reps sr
on sr.id = a.sales_rep_id
join region r
on r.id = sr.region_id
group by r.name
order by count(a.id) desc;
```



The screenshot shows a database query tool interface. At the top, there's a tab labeled 'region 1'. Below it, a SQL query is entered: `select r."name" region, count(a.id) num_of_acc`. The results are displayed in a table with two columns: 'region' and 'num_of_account'. The regions listed are Northeast (107), West (101), Southeast (96), and Midwest (48). The table is sorted by 'num_of_account' in descending order.

region	num_of_account
Northeast	107
West	101
Southeast	96
Midwest	48

----Identify how many sales representatives are allocated for each region

```
select r.name region, count(sr.id) num_of_representative from sales_reps sr
left join region r
on r.id = sr.region_id
group by region
order by count(sr.id) desc;
```

region 1 X

select r.name region, count(sr.id) num_of_repr Enter a SQL expression to filter results (use Ctrl+Space)

Grid	A-Z region	123 num_of_representative
1	Northeast	21
2	Southeast	10
3	West	10
4	Midwest	9
5	[NULL]	1
6	International	1

Refresh Save Cancel Export data 200 6 Cells: 1

----Identify the sales performance for each region

```
select sr.region_id region, sum(total) total_quantity ,round(sum(total_amt_usd))
total_amt_usd
from orders o
JOIN accounts a
ON a.id = o.account_id
join sales_reps sr
on sr.id=a.sales_rep_id
group by sr.region_id
order by total_amt_usd DESC;
```

sales_reps 1 X

select sr.region_id region, sum(total) total_qua Enter a SQL expression to filter results (use Ctrl+Space)

Grid	123 region	123 total_quantity	123 total_amt_usd
1	1	1,230,378	7,744,405
2	3	1,035,005	6,458,497
3	4	927,532	5,925,123
4	2	482,850	3,013,487

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----Identify accounts and sales representative

```
select a.sales_rep_id ,sum(o.standard_amt_usd)
from accounts a
join orders o
on a.id =o.account_id
group by a.sales_rep_id ;
```

accounts 1 X

select a.sales_rep_id,sum(o.standard_amt_us | Enter a SQL expression to filter results (use Ctrl+Space)

	123 sales_rep_id	123 sum
1	321,940	234,699.66
2	321,910	224,220.66
3	321,720	85,204.25
4	321,670	14,840.26
5	321,610	141,765.9
6	321,830	306,191.39
7	321,890	335,432.79
8	321,590	160,687.98

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----Identify each sales representative allocated to how many accounts

```
select sales_rep_id ,count(id)
from accounts a
group by sales_rep_id ;
```

accounts 1 X

select sales_rep_id ,count(id) from accounts a | Enter a SQL expression to filter results (use Ctrl+Space)

	123 sales_rep_id	123 count
1	321,940	10
2	321,910	10
3	321,720	5
4	321,670	3
5	321,610	3
6	321,830	11
7	321,890	7
8	321,590	6

Refresh Save Cancel Export data 200 50 Cells: 1

--4b

----number of reps per region

```
select r.name as region, count(sr.id) as num_sales_reps
from sales_reps sr
join region r on sr.region_id = r.id
group by r.name
order by num_sales_reps desc;
```

region 1 X

select r.name as region, count(sr.id) as num_sales_reps

Grid	A-Z region	123 num_sales_reps
1	Northeast	21
2	Southeast	10
3	West	10
4	Midwest	9
5	International	1

Refresh Save Cancel Export data 200 5 Cells: 1

----total orders and revenue per region

```
select r.name, sum(o.total) as qty_sold, sum(total_amt_usd) as total_revenue
from orders o
join accounts a on o.account_id = a.id
join sales_reps sr on a.sales_rep_id = sr.id
join region r on sr.region_id = r.id
group by r.name
order by sum(total_amt_usd) desc;
```

region 1 X

select r.name, sum(o.total) as qty_sold, sum(total_amt_usd) as total_revenue

Grid	A-Z name	123 qty_sold	123 total_revenue
1	Northeast	1,230,378	7,744,405.36
2	Southeast	1,035,005	6,458,497
3	West	927,532	5,925,122.96
4	Midwest	482,850	3,013,486.51

Refresh Save Cancel Export data 200 4 Cells: 1

--Q5

Marketing Strategies

--5a

----To segregate:

```
select 'Food' as industry_category,
       count(*) as customer_count
from accounts
where name ilike '%Food%'
       or name ilike '%Restaurant%'
```

```

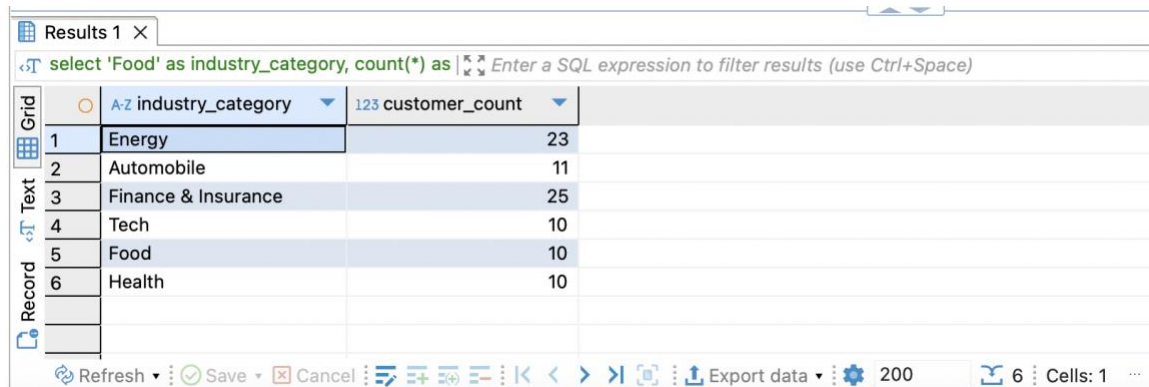
    or name ilike '%Cafe%'
    or name ilike '%Catering%'
    or name ilike '%Soup%'
    or name ilike '%Market%'
union
select 'Finance & Insurance' as industry_category,
       count(*) as customer_count
from accounts
where name ilike '%financ%'
    or name ilike '%Bank%'
    or name ilike '%Capital%'
    or name ilike '%Investment%'
    or name ilike '%Insurance%'
    or name ilike '%Insure%'
    or name ilike '%Policy%'
    or name ilike '%Life%'
union
select 'Energy' as industry_category,
       count(*) as customer_count
from accounts
where name ilike '%energy%'
    or name ilike '%power%'
    or name ilike '%electric%'
    or name ilike '%oil%'
    or name ilike '%gas%'
    or name ilike '%petro%'
    or name ilike '%renewable%'
union
select 'Automobile' as industry_category,
       count(*) as customer_count
from accounts
where name ilike '%auto%'
    or name ilike '%motor%'
union

```

```

select 'Tech' as industry_category,
       count(*) as customer_count
from accounts
where name ilike '%tech%'
       or name ilike '%software%'
       or name ilike '%digital%'
       or name ilike '%communication%'
       or name ilike '%solution%'
       or name ilike '%system%'
union
select 'Health' as industry_category,
       count(*) as customer_count
from accounts
where name ilike '%health%'
       or name ilike '%hospital%'
       or name ilike '%pharma%';

```



Results 1 x

select 'Food' as industry_category, count(*) as customer_count

	A-Z industry_category	123 customer_count
1	Energy	23
2	Automobile	11
3	Finance & Insurance	25
4	Tech	10
5	Food	10
6	Health	10

Refresh Save Cancel Export data 200 6 Cells: 1

-- To get the final count of customers and then select top 2:

```

select industry_category,
       count(distinct orders.id) as total_orders
from (
  select id, 'Food' as industry_category
  from accounts
  where name ilike '%Food%'
        or name ilike '%Restaurant%'
        or name ilike '%Cafe%'
        or name ilike '%Catering%'

```

```
    or name ilike '%Soup%'
    or name ilike '%Market%'
union all
select id, 'Finance & Insurance' as industry_category
from accounts
where name ilike '%financ%'
    or name ilike '%Bank%'
    or name ilike '%Capital%'
    or name ilike '%Investment%'
    or name ilike '%Insurance%'
    or name ilike '%Insure%'
    or name ilike '%Policy%'
    or name ilike '%Life%'
union all
select id, 'Energy' as industry_category
from accounts
where name ilike '%energy%'
    or name ilike '%power%'
    or name ilike '%electric%'
    or name ilike '%oil%'
    or name ilike '%gas%'
    or name ilike '%petro%'
    or name ilike '%renewable%'
union all
select id, 'Automobile' as industry_category
from accounts
where name ilike '%auto%'
    or name ilike '%motor%'
union all
select id, 'Tech' as industry_category
from accounts
where name ilike '%tech%'
    or name ilike '%software%'
    or name ilike '%digital%'
```

```

or name ilike '%communication%'
or name ilike '%solution%'
or name ilike '%system%'
union all
select id, 'Health' as industry_category
from accounts
where name ilike '%health%'
      or name ilike '%hospital%'
      or name ilike '%pharma%'
) as categorized_companies
join orders on orders.account_id = categorized_companies.id
group by industry_category
order by total_orders desc;

```

Results 1 X

select *industry_category*, count(distinct orders) Enter a SQL expression to filter results (use Ctrl+Space)

	A-Z <i>industry_category</i>	123 <i>total_orders</i>
1	Finance & Insurance	471
2	Energy	417
3	Tech	291
4	Automobile	180
5	Food	149
6	Health	90

Refresh Save Cancel Export data 200 6 Cells: 1

--5b

```

select q.region_id, we.channel, count(q.region_id) as counts
from web_events we
left join (select a.id, a.name, a.sales_rep_id, sr.region_id
            from accounts a
            left join sales_reps sr
            on a.sales_rep_id = sr.id) q
on we.account_id = q.id
group by we.channel, q.region_id
order by q.region_id, counts asc

```

